

Feature Scaling Methods

There are several common techniques to scale features so they fall within a similar range, which helps gradient descent converge faster.

1. Max Scaling

This is one of the simplest methods. You just divide each feature by its maximum value in the training set.

- **Formula:** $x_{scaled} = \frac{x_{original}}{max(x)}$
 - **Resulting Range:** The scaled feature will typically range from just above 0 to 1.
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2. Mean Normalization

This method centers the feature values around zero.

- **Process:**
 1. Calculate the **mean** (average), μ , of the feature in the training set.
 2. Calculate the **range** of the feature (i.e., $max(x) - min(x)$).
 - **Formula:** $x_{scaled} = \frac{x_{original} - \mu}{max(x) - min(x)}$
 - **Resulting Range:** The scaled feature will be centered around 0.
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3. Z-score Normalization

This is a very common technique that scales features based on their mean and standard deviation.

- **Process:**
 1. Calculate the **mean**, μ , of the feature.
 2. Calculate the **standard deviation**, σ , of the feature.
 - **Formula:** $x_{scaled} = \frac{x_{original} - \mu}{\sigma}$
 - **Resulting Range:** The scaled feature will have a mean of 0 and a standard deviation of 1.
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Rule of Thumb

The general goal of feature scaling is to get your features into a range of roughly **-1 to +1**.

- **Acceptable Ranges:** Ranges like **-3 to +3** or **-0.3 to +0.3** are perfectly fine and don't necessarily need scaling.

- **Ranges to Scale:** You should consider scaling features with very large or very small ranges, as these can slow down gradient descent.
 - **Too Large:** A range like **-100 to +100**.
 - **Too Small:** A range like **-0.001 to +0.001**.

When in doubt, it's almost never harmful to perform feature scaling.